Center-Aware Global-to-Local Modeling for Patch-Based HSI–LiDAR Classification

Mingwan Li, Sheng Fang, Zhe Li, *Member, IEEE*, Jianli Zhao, and Ruisen Huang

***Abstract*—Hyperspectral imagery (HSI) and light detection and ranging (LiDAR) provide complementary spectral and elevation cues for fine-grained land-cover classification, yet accurate pixel-wise fusion remains challenging in heterogeneous scenes. Most deep HSI–LiDAR classifiers follow a center-supervised patch-based setting, where supervision is defined on the center pixel while predictions are inferred from a context-enhanced patch representation. Under this setting, mixed semantics within a patch induce *heterogeneous-patch ambiguity*: contextual pixels exert uneven influence on the center-pixel decision, and indiscriminate context aggregation can dilute center-consistent evidence in mixed and boundary regions. To address this issue, we propose a center-aware global-to-local refinement framework that explicitly regulates how contextual information is organized and accumulated under center supervision. First, a lightweight cross-modal channel alignment module fuses HSI and LiDAR features into a unified representation while preserving the spatial layout. Second, we introduce a center-aware global regulation module built on Vision Mamba, equipped with a dual-direction Spiral Scan that orders tokens in a periphery-to-center manner. This design induces a structured information flow that progressively consolidates center-relevant semantics under heterogeneous neighborhoods. Finally, a lightweight spatial–spectral refinement module (SSRM) refines discriminative local details within the globally regulated feature space by recovering boundary-sensitive structures and recalibrating channel responses. Extensive experiments on three public benchmarks (Houston2013, MUUFL, and Augsburg) demonstrate that the proposed method consistently outperforms representative local neighborhood modeling, within-patch global interaction, and local–global hybrid approaches. The code is available at https://github.com/lmwdhr/ViT–CNN**

***Index Terms*—Center-pixel classification, global-to-local refinement, heterogeneous-patch ambiguity, hyperspectral imagery (HSI), land-cover classification, light detection and ranging (LiDAR), Mamba, multimodal fusion, spatial–spectral refinement, Spiral Scan, vision state-space models (SSMs).**

I. Introduction

HYPERSPECTRAL imagery (HSI) and light detection and ranging (LiDAR) provide highly complementary cues for fine-grained land-cover classification. HSI records material-specific spectral signatures, whereas LiDAR supplies elevation and geometric information that is particularly valuable when categories exhibit similar spectral characteristics but distinct 3-D structures. The benefit of combining these modalities has been validated in joint classification studies, where LiDAR-derived elevation and structural cues help resolve ambiguities that are difficult to address using spectral data alone [1], [2], [3]. Consequently, HSI–LiDAR fusion has been widely adopted for detailed land-cover mapping and complex scene interpretation [2], [3], [4], [5], [6], [7], [8], [9].

Despite these advantages, accurate *pixel-wise* multimodal classification remains challenging in highly heterogeneous scenes. Modern deep learning frameworks typically perform pixel-wise classification by extracting a local patch around each labeled pixel and predicting the class of the *center pixel* [10], [11], [12]. This patch-based paradigm is prevalent because it injects spatial context, and numerous studies have shown that jointly exploiting the spatial–spectral information of HSI together with LiDAR-derived elevation and geometric cues consistently improves classification accuracy [13], [14], [15].

However, patch-based modeling also introduces an intrinsic difficulty: a local patch almost inevitably contains multiple semantic categories, especially near boundaries or in mixed regions. We refer to such neighborhoods as *heterogeneous patches*, where pixels from different land-cover classes coexist within a single patch. In this setting, supervision is defined only on the center pixel, while surrounding pixels serve as contextual evidence for prediction, as illustrated in Fig. 1(a). Under heterogeneous patches, contextual pixels can exert highly nonuniform influence on the center-pixel prediction. Pixels belonging to the same class as the center often provide supportive evidence, whereas pixels from other classes may introduce signals that are not directly aligned with the center-supervised objective, as shown in Fig. 1(b). When such heterogeneous context is aggregated in a generic manner, class-consistent evidence associated with the center pixel can be diluted by competing contextual signals, leading to confusion in boundary or mixed areas [10], [16]. We term this phenomenon *heterogeneous-patch ambiguity*. To make this phenomenon measurable, we introduce a center-label inconsistency ratio that quantifies the proportion of labeled contextual pixels whose semantic labels differ from that of the center pixel. A larger inconsistency ratio indicates stronger heterogeneous-patch ambiguity and greater potential interference from center-irrelevant context. Detailed definitions and heterogeneous-patch ambiguity-level evaluation protocols are provided in Section IV.

This center-supervised perspective suggests that the key issue is not merely whether contextual information is used,

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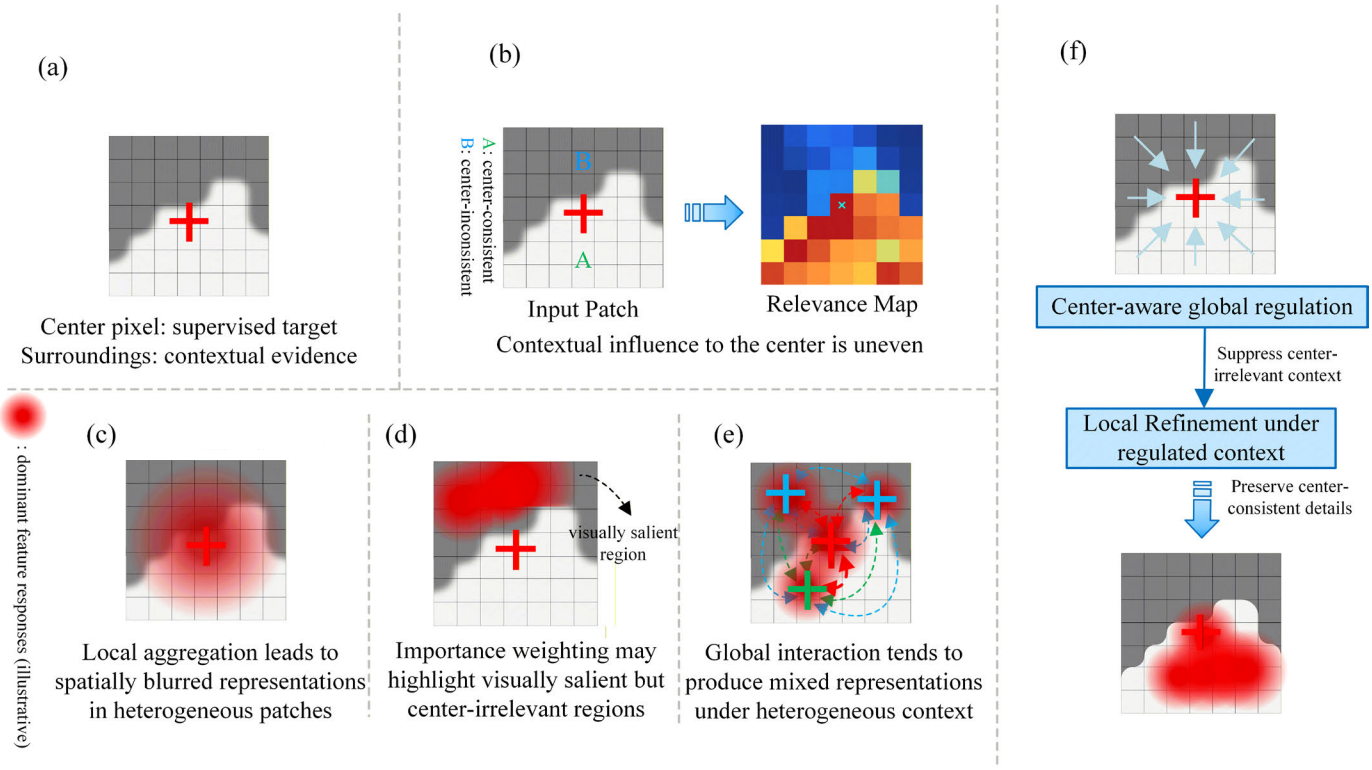


Fig. 1. Motivation and principle of center-aware global-to-local modeling for patch-based HSI–LiDAR classification. The proposed center-aware global-to-local refinement effectively suppresses center-irrelevant context while preserving center-consistent details under heterogeneous patches. (a) Centered-supervised patch-based classification. (b) Uneven contextual influence under heterogeneous patches. (c) Local aggregation. (d) Importance weighting. (e) Global interaction. (f) Center aware global-to-local refinement principle.

but how heterogeneous context is organized and regulated with respect to the center target. Accordingly, existing patch-based HSI–LiDAR classification methods can be broadly reviewed according to their context-organization mechanisms, including local context aggregation, importance-weighted context modulation, and within-patch global context interaction.

Local context aggregation methods, typically implemented with CNN-based operators, exploit neighboring pixels through fixed local receptive fields [3], [17]. Their main advantage lies in preserving local spatial continuity and fine-grained spatial–spectral patterns. However, such aggregation is usually performed directly on raw or weakly organized neighborhood features. Under heterogeneous patches, pixels from different semantic classes may be progressively mixed by local convolutions, making center-consistent evidence vulnerable to dilution by surrounding center-irrelevant regions.

Importance-weighted context modulation methods introduce spatial, channel, or cross-modal weighting mechanisms to adaptively recalibrate feature responses [18], [19], [20], [21]. Compared with fixed local aggregation, these methods can emphasize informative regions or channels. Nevertheless, the weighting process is often applied after local feature extraction and mainly reflects feature saliency or modality importance. It does not explicitly organize contextual pixels according to their contribution to the center-supervised target. Therefore, visually salient but center-inconsistent regions may still receive high responses in mixed or boundary areas.

Within-patch global context interaction methods further allow long-range information exchange among patch elements through Transformer-based token mixing, graph-based relational propagation, or state-space sequence modeling [5], [6], [7], [8], [22], [23], [24], [25], [26], [27]. Their essential difference from local aggregation is that each patch element can interact with distant positions, thereby providing a broader contextual view. However, most global interaction mechanisms are designed as generic token- or graph-mixing operations. They usually do not consider the asymmetric center-supervised setting, where only the center pixel defines the label while other pixels provide auxiliary context with uneven relevance.

Overall, the essential differences among these strategies lie in how they organize contextual information: local aggregation relies on fixed neighborhood accumulation, importance weighting recalibrates feature responses, and global interaction enables all-token relational modeling. However, none of them explicitly addresses the uneven contextual influence caused by heterogeneous patches under center supervision.

Motivated by this observation, we propose a *center-aware global-to-local* refinement framework for HSI–LiDAR classification, as conceptually illustrated in Fig. 1(f). The key idea is to explicitly address the uneven influence of heterogeneous context by first regulating patch-level contextual information at a global scale and then refining discriminative local details within this regulated context. Specifically, in the global stage, patch-level context is first regulated in a center-oriented manner to progressively mitigate the interference

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caused by uneven contextual influence under heterogeneous patches, yielding a stable and discriminative patch representation. This global regulation stage operates on the entire patch and establishes a coherent semantic abstraction with an effectively large receptive field, in which center-relevant contextual cues are consolidated while semantically irrelevant responses are suppressed.

In the local stage, spatial–spectral refinement is subsequently performed *within this globally regulated feature space*. As a result, local operators no longer act on raw or weakly organized neighborhood features, but instead refine boundary-sensitive structures and spectral coherence under top-down semantic guidance. This global-to-local ordering can be interpreted as an implicit form of semantic conditioning, where the globally regulated representation constrains and guides local refinement, facilitating more focused boundary enhancement and more reliable center-pixel prediction under heterogeneous neighborhoods.

The main contributions of this work are summarized as follows.

1) *Problem Formulation and Perspective:* We identify *heterogeneous-patch ambiguity* as a fundamental challenge in patch-based HSI–LiDAR classification, arising from the uneven influence of heterogeneous contextual pixels under center supervision.
2) *Center-Aware Global-to-Local Framework:* We propose a center-aware global-to-local refinement framework that explicitly regulates patch-level contextual modeling in a two-stage manner, addressing uneven contextual influence under center supervision.
3) *Center-Oriented Global Regulation:* We develop a center-aware global regulation module based on Vision Mamba with a dual-direction Spiral Scan, which induces a structured accumulation of contextual information with respect to the center position under heterogeneous patches.
4) *Local Spatial–Spectral Refinement:* We introduce a lightweight spatial–spectral refinement module to enhance fine-grained spatial structures and spectral coherence within the globally regulated patch representation.
5) *Extensive Evaluation:* Comprehensive experiments on three public datasets demonstrate that the proposed framework consistently outperforms state-of-the-art HSI–LiDAR classification methods, with further ablation studies validating the contribution of each component.

II. Related Work

A. Local Neighborhood Modeling for Patch-Based HSI–LiDAR Classification

Most contemporary HSI–LiDAR classification methods adopt a center-supervised patch-based setting, where a local neighborhood is extracted around each labeled pixel and the class of the center pixel is inferred from neighborhood evidence. A dominant line of research focuses on *local* spatial–spectral representation learning with convolutional backbones that exploit neighborhood continuity.

Early CNN-based multimodal architectures model local neighborhoods by combining HSI spatial–spectral cues with LiDAR elevation cues, aiming to alleviate spectral ambiguity via complementary elevation and geometry information. Beyond early multibranch CNN fusion recipes [3], [11], [17], [28], subsequent studies improve local modeling from different angles. For instance, Lu et al. [29] introduce coupled adversarial learning to enhance cross-modal feature alignment, and DSHFNet [30] refines local representations via dynamic-scale fusion and attention-guided integration.

Overall, these local neighborhood modeling methods are effective and computationally practical, yet they primarily accumulate contextual information without explicitly regulating how different regions contribute to the center-pixel decision. Under heterogeneous neighborhoods, such unregulated aggregation can allow semantically irrelevant or competing regions to dominate, thereby exacerbating ambiguity in the center-supervised patch representation.

B. Within-Patch Global Interaction and State-Space Scan Mechanisms

Beyond local neighborhood extraction, another line of work investigates *within-patch global interaction*, where each element in a patch can interact with all other positions. Transformer-based models implement this idea via self-attention with data-dependent weights. For example, LIIT [31] couples HSI and LiDAR tokens within a neighborhood, and CHLAT [21] further strengthens cross-modal exchange via attention. Graph-based approaches provide an alternative by constructing pixel or superpixel graphs and propagating information along learned relations [26], [27]. More recently, vision state-space models (SSMs) (e.g., Mamba variants) offer another global-mixing mechanism with linear complexity while maintaining strong long-range dependency modeling ability [5], [6], [7], [8], [32]. These methods demonstrate the growing potential of SSM in multimodal remote sensing classification.

A key distinction of Vision Mamba is that 2-D features are converted into 1-D sequences for state-space processing, making the *scan mechanism* an important design factor: different scan orders impose different inductive biases on how information propagates and aggregates across spatial positions, as illustrated in Fig. 2. Vim [33] introduces forward and reverse scanning, VMamba [25] extends this idea with 2-D scanning, and subsequent works explore richer strategies such as FRR scanning in RSMamba [34], omnidirectional scans [35], and task-tailored scans for hyperspectral imagery (e.g., SpectralMamba [36]).

These studies highlight that scan design can materially influence what dependencies an SSM captures. However, most scan strategies target generic visual modeling and do not explicitly account for the asymmetry of center-supervised patch-based HSI–LiDAR classification, where only the center pixel defines supervision and other pixels provide auxiliary context with nonuniform relevance. This motivates center-aware scan designs that bias information flow toward

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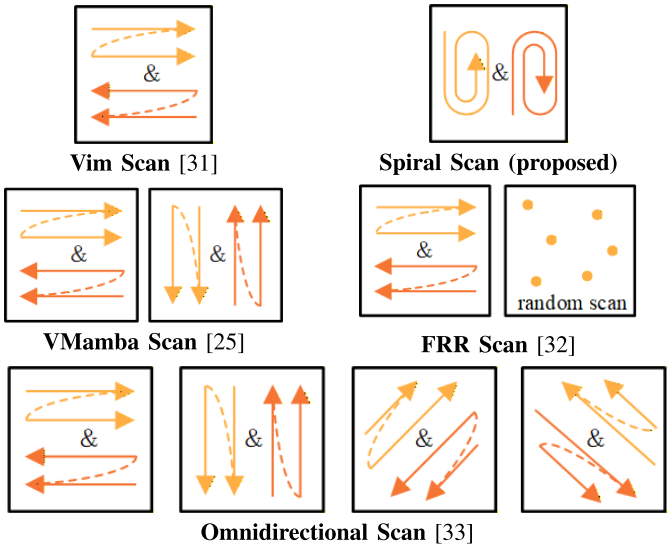


Fig. 2. Representative scanning strategies in Vision Mamba and the proposed Spiral Scan.

consolidating center-relevant semantics under heterogeneous neighborhoods; in our work, a spiral-like periphery-to-center scan (with dual directions) provides a globally regulated representation for subsequent local refinement.

C. Local–Global Hybrid Designs and Global-to-Local Refinement

Local operators and global interaction are complementary: local modeling preserves fine-grained structures and boundary-sensitive patterns, while global interaction facilitates holistic semantic interpretation and long-range context integration. Accordingly, hybrid designs that combine local and global modeling have attracted extensive attention in computer vision and remote sensing.

One strategy injects convolutions into Transformer blocks to introduce spatial inductive biases and improve efficiency, as exemplified by CoAtNet [37] and MobileViT [38]. Beyond intrablock hybridization, more explicit hybrid frameworks combine local and global modeling stages in a sequential or parallel manner. Serial local-to-global pipelines (e.g., CNN → ViT) are widely adopted in large-input scenarios; for instance, Early Conv [39] employs convolutional embedding to preserve local details before applying Transformer-based global interaction. HSI–LiDAR joint classification follows similar design principles by converting multimodal neighborhood features into token-like representations and applying global interaction mechanisms. Representative examples include token-based Transformer fusion [40], multiscale CNN—Vision Transformer (ViT) hybrids [41], and parallel spectral–spatial designs where convolutional branches extract local features and Transformers capture long-range dependencies [42]. Parallel hybrid architectures further fuse local and global branches through coupling modules, such as Conformer [43]. Importantly, recent studies have also shown that local-to-global modeling can be achieved without explicit convolutional operators, for example through graph-based or transformer-only mechanisms that rely on relational propagation or global attention, such as GraphGST [44]. These approaches further indicate that the effectiveness of hybrid modeling is not tied to a specific operator choice (CNN versus graph versus Transformer), but rather to how local and global information are organized and integrated.

Despite their strong performance, existing hybrid methods are predominantly designed under a local-to-global

Algorithm 1 Center-Aware Global-to-Local Modeling With Spiral Scan

Require: HSI patch $\mathbf{H} \in \mathbb{R}^{h\times w\times C_H}$, LiDAR patch $\mathbf{L} \in \mathbb{R}^{h\times w\times C_L}$ (odd h, w)
Require: Predefined spiral indices $\pi_{\text{cw}}, \pi_{\text{ccw}} \in \{1,\dots,N\}^N$ for $N = h \cdot w$
Require: Mamba stack $\{\text{MambaLayer}_\ell\}_{\ell=1}^{L}$, learnable direction weights $\mathbf{w} \in \mathbb{R}^2$
Ensure: Refined feature map $\mathbf{F}_{\text{out}} \in \mathbb{R}^{h\times w\times d}$ (for center-pixel classification)

function PERMUTE($\mathbf{X}, \pi$) ▷$\mathbf{X} \in \mathbb{R}^{B\times N\times d}$
return $\mathbf{X}[:, \pi, :]$
end function
function INVPERMUTE($\mathbf{X}, \pi$)
  $\pi^{-1} \leftarrow \text{Argsort}(\pi)$
  return $\mathbf{X}[:, \pi^{-1}, :]$
end function
(A) Channel Fusion (token-wise)
$\mathbf{H}_{\text{seq}} \leftarrow \text{Flatten}(\mathbf{H}) \in \mathbb{R}^{N\times C_H}$, $\mathbf{L}_{\text{seq}} \leftarrow \text{Flatten}(\mathbf{L}) \in \mathbb{R}^{N\times C_L}$
$\mathbf{Z}_H \leftarrow \phi^H_{\text{mlp}}(\mathbf{H}_{\text{seq}}) \in \mathbb{R}^{N\times d_H}$, $\mathbf{Z}_L \leftarrow \phi^L_{\text{mlp}}(\mathbf{L}_{\text{seq}}) \in \mathbb{R}^{N\times d_L}$
$\mathbf{Z} \leftarrow \phi^C_{\text{mlp}}(\text{Concat}_c(\mathbf{Z}_H, \mathbf{Z}_L)) \in \mathbb{R}^{N\times d}$
$\mathbf{Z}' \leftarrow \phi_{\text{se}}(\mathbf{Z}) \in \mathbb{R}^{N\times d}$
$\mathbf{X} \leftarrow \mathbf{Z}'$ and reshape to $\mathbf{F} \in \mathbb{R}^{h\times w\times d}$
(B) Spiral Scan + Mamba Global Regulation
$\mathbf{X} \leftarrow \text{Flatten}(\mathbf{F}) \in \mathbb{R}^{1\times N\times d}$ ▷batch omitted for clarity
for $\ell = 1$ to L **do**
  $\mathbf{R} \leftarrow \mathbf{X}$ ▷residual
  $\mathbf{S}_{\text{cw}} \leftarrow \text{PERMUTE}\mathbf{X}, \pi_{\text{cw}}$
  $\mathbf{S}_{\text{ccw}} \leftarrow \text{PERMUTE}\mathbf{X}, \pi_{\text{ccw}}$
  $\mathbf{S} \leftarrow \text{Concat}_b(\mathbf{S}_{\text{cw}}, \mathbf{S}_{\text{ccw}})$ ▷stack along batch: $2B \times N \times d$
  $\mathbf{S} \leftarrow \text{PreNorm}(\mathbf{S})$
  $\hat{\mathbf{S}} \leftarrow \text{MambaLayer}_\ell(\mathbf{S})$
  split $\hat{\mathbf{S}}$ into $\hat{\mathbf{S}}_{\text{cw}}, \hat{\mathbf{S}}_{\text{ccw}}$
  $\hat{\mathbf{X}}_{\text{cw}} \leftarrow \text{INVPERMUTE}(\hat{\mathbf{S}}_{\text{cw}}, \pi_{\text{cw}})$
  $\hat{\mathbf{X}}_{\text{ccw}} \leftarrow \text{INVPERMUTE}(\hat{\mathbf{S}}_{\text{ccw}}, \pi_{\text{ccw}})$
  $\boldsymbol{\alpha} \leftarrow \text{Softmax}(\mathbf{w})$ ▷$\boldsymbol{\alpha} = [\alpha_1, \alpha_2]$
  $\mathbf{X} \leftarrow \alpha_1\hat{\mathbf{X}}_{\text{cw}} + \alpha_2\hat{\mathbf{X}}_{\text{ccw}}$
  $\mathbf{X} \leftarrow \mathbf{R} + \mathbf{X}$ ▷residual connection
end for
$\mathbf{F}_{\text{inter}} \leftarrow \text{Unflatten}(\mathbf{X}) \in \mathbb{R}^{h\times w\times d}$
(C) SSRM Local Refinement
$\mathbf{F}_{\text{spa}} \leftarrow \text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(\mathbf{F}_{\text{inter}})))$
$\mathbf{F}_{\text{spe}} \leftarrow \text{ReLU}(\text{LN}(\text{Conv}_{1\times1}(\mathbf{F}_{\text{inter}})))$
$\mathbf{F}_{\text{out}} \leftarrow \mathbf{F}_{\text{spa}} + \mathbf{F}_{\text{spe}}$
return $\mathbf{F}_{\text{out}}$

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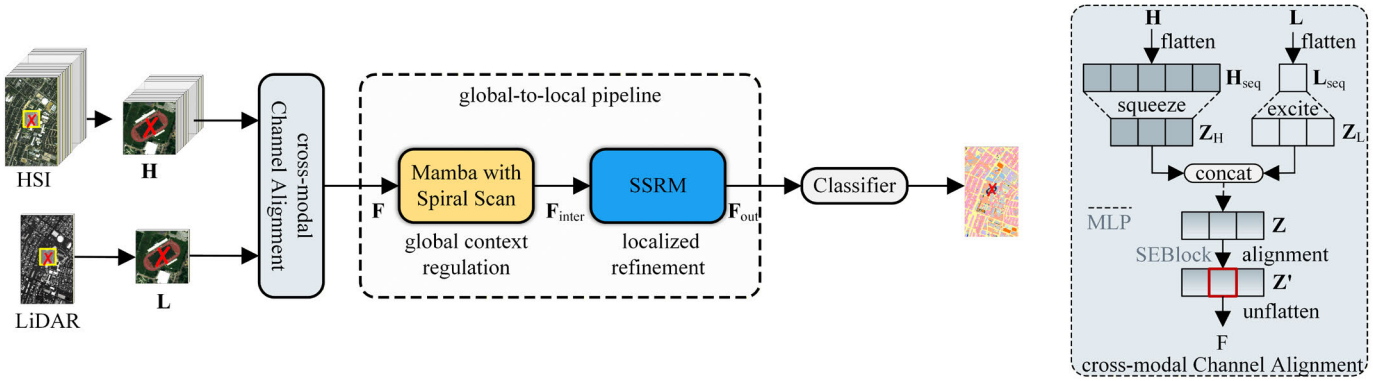


Fig. 3. Overall architecture of the proposed center-aware global-to-local refinement framework. HSI and LiDAR inputs are first aligned via cross-modal channel fusion, followed by center-targeted global regulation using Vision Mamba with Spiral Scan and subsequent SSRM for center-pixel classification.

organization paradigm. In the center-supervised patch-based HSI–LiDAR setting, such generic combinations do not explicitly account for the asymmetric supervision defined by the center pixel. If local aggregation is performed before semantic stabilization, context from semantically distant regions may interfere with the center-pixel representation; conversely, if global interaction is applied without center-oriented regulation, heterogeneous context may still be indiscriminately mixed.

These observations suggest that the combination strategy itself, rather than the specific choice of operators, plays a critical role. Rather than simply stacking local and global modules, an effective design should first regulate within-patch contextual information at a global stage in a center-aware manner, and then perform local refinement within a stabilized semantic context. Such a global-to-local refinement strategy also makes scan-path ablations meaningful, as different scan orders directly influence how contextual evidence is accumulated toward the center position prior to local enhancement.

III. Methodology

As shown in Fig. 3, we adopt a patch-based setting where each input neighborhood is labeled by its center pixel, and the network predicts the class based on a context-enhanced patch representation. To mitigate heterogeneous-patch ambiguity, our framework follows a global-to-local refinement.

We first perform a lightweight cross-modal channel alignment to obtain a unified feature map $\mathbf{F}$ while preserving the spatial layout. Then, a center-aware Mamba encoder with Spiral Scan conducts global context regulation by progressively consolidating center-relevant semantics under heterogeneous neighborhoods. Finally, the proposed SSRM performs local refinement on the globally regulated features to recover boundary-sensitive and class-consistent local structures that support the center-defined label, improving robustness in heterogeneous neighborhoods. The overall global-to-local forward process is summarized in Algorithm 1.

The remainder of this section is organized as follows. Section III-A details the channel alignment design. Section III-B presents Vision Mamba with the proposed Spiral Scan. Section III-C introduces SSRM and the center-supervised patch-level classifier.

A. Cross-Modal Channel Alignment

Given an HSI patch $\mathbf{H} \in \mathbb{R}^{h\times w\times C_H}$ and a LiDAR patch $\mathbf{L} \in \mathbb{R}^{h\times w\times C_L}$ (typically C_L=1), we first convert them into pixel-level token sequences. We apply modality-specific projections to compress HSI channels and enrich LiDAR embeddings, and then align and mix the two modalities along the channel dimension

$$\begin{aligned}
\mathbf{H}_{\text{seq}} &= \text{Flatten}\,(\mathbf{H}) \in \mathbb{R}^{N\times C_H} \\
\mathbf{L}_{\text{seq}} &= \text{Flatten}\,(\mathbf{L}) \in \mathbb{R}^{N\times C_L} \\
\mathbf{Z}_H &= \phi^{H}_{\text{mlp}}\left(\mathbf{H}_{\text{seq}}\right) \in \mathbb{R}^{N\times d_H} \\
\mathbf{Z}_L &= \phi^{L}_{\text{mlp}}\left(\mathbf{L}_{\text{seq}}\right) \in \mathbb{R}^{N\times d_L} \\
\mathbf{Z} &= \phi^{C}_{\text{mlp}}\left(\text{Concat}_c\left(\mathbf{Z}_H, \mathbf{Z}_L\right)\right) \in \mathbb{R}^{N\times d} \\
\mathbf{Z}' &= \phi_{\text{se}}\left(\mathbf{Z}\right) \in \mathbb{R}^{N\times d} \\
\mathbf{F} &= \text{Unflatten}\left(\mathbf{Z}'\right) \in \mathbb{R}^{h\times w\times d}
\end{aligned} \tag{1}$$

where $N = h \times w$ is the number of spatial tokens, $\text{Concat}_c(\cdot)$ denotes channel-wise concatenation, and $\phi_{\text{se}}(\cdot)$ is an SE block [45].

This design serves as an early channel-alignment stage before Mamba-based global regulation. Since HSI contains high-dimensional spectral responses with potential redundancy, whereas LiDAR provides low-dimensional but complementary elevation cues, direct concatenation may produce unbalanced multimodal tokens. The modality-specific MLPs first project the two modalities into a more balanced latent space, and the following cross-modal MLP and SE block further mix and recalibrate informative spectral–elevation channels. Meanwhile, the 2-D spatial layout is preserved, enabling the subsequent Mamba encoder to regulate already-fused multimodal pixel tokens in a center-aware manner.

B. Center-Aware Vision Mamba With Dual-Direction Spiral Scan

Given the aligned feature map $\mathbf{F} \in \mathbb{R}^{h\times w\times d}$, the global stage applies Vision Mamba to model within-patch dependencies. Vision Mamba performs order-sensitive SSM, where the scan order directly governs how contextual information is recursively accumulated. Let L denote the number of S6 blocks, d denote the embedding dimension, and $N = h \times w$ denote the

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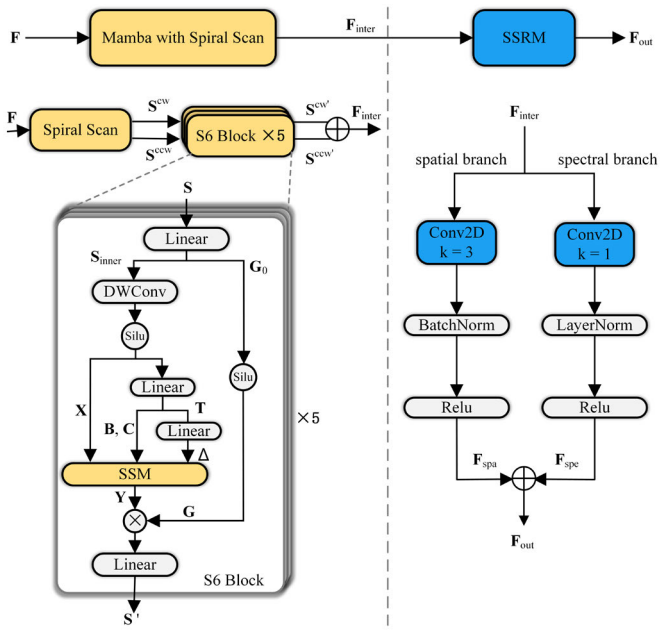


Fig. 4. Implementation of the proposed global-to-local refinement pipeline. Mamba with Spiral Scan conducts center-targeted global regulation, and SSRM refines the regulated features through spatial–spectral local enhancement.

sequence length. Unless otherwise specified, we set $L = 5$ and $d = 64$ in the main experiments, according to the sensitivity analysis reported in Section IV-B4.

As shown in Fig. 4, the proposed global regulation stage generates two spiral-ordered sequences (clockwise and counterclockwise), processes them with the same Mamba encoder, and fuses the two directions with a learnable softmax gate. By design, the center position is placed near the *tail* of each spiral sequence; due to the SSM recurrence, its hidden state aggregates contextual evidence accumulated from earlier (periphery) tokens, thereby stabilizing center semantics, as visually illustrated in Fig. 5.

1) Spiral Scan (Dual-Direction Periphery-to-Center Ordering): For clarity and reproducibility, the Spiral Scan procedure and its integration with the Mamba layers are provided in Algorithm 1. Patch-based HSI–LiDAR classification often adopts an odd-sized neighborhood so that a unique center pixel exists.

Let the fused feature map be $\mathbf{F} \in \mathbb{R}^{h\times w\times d}$ with odd h, w, and $N = h \times w$. We flatten $\mathbf{F}$ into a token sequence $\mathbf{X} = \text{Flatten}(\mathbf{F}) \in \mathbb{R}^{N\times d}$.

a) Predefined spiral traversal: As conceptually illustrated by the Spiral Scan in Fig. 2, we predefine two deterministic spiral traversal indices according to the patch size: a clockwise spiral ($\pi_{\text{sp}}^{\text{cw}}$) and a counterclockwise spiral ($\pi_{\text{sp}}^{\text{ccw}}$). Both indices visit tokens ring by ring from the patch boundary toward the center, but use opposite within-ring visiting directions. Therefore, they are fixed permutations determined only by the patch size and traversal direction, and can be generated once before training. Using these indices, we generate two spiral-ordered sequences by permutation

$$\mathbf{S}^{\text{cw}} = \text{Permute}_{\pi_{\text{sp}}^{\text{cw}}}(\mathbf{X})$$
$$\mathbf{S}^{\text{ccw}} = \text{Permute}_{\pi_{\text{sp}}^{\text{ccw}}}(\mathbf{X}) \tag{2}$$

where $\mathbf{S}^{\text{cw}}, \mathbf{S}^{\text{ccw}} \in \mathbb{R}^{N\times d}$. Since the traversal proceeds from the boundary to the center, the center token appears near the tail of each sequence. This ordering is aligned with center-defined patch labeling (see Fig. 5). Since the state-space recurrence propagates information along the scan sequence, the periphery-to-center traversal shapes how contextual evidence is integrated into token representations. After global pooling, this yields a patch-level representation that is less dominated by semantically distant regions under heterogeneous neighborhoods.

b) Order-sensitive global regulation and fusion: We feed the two spiral-ordered sequences into an identical S6 stack $\Phi_{\text{S6}}^{(L)}(\cdot)$ with shared parameters

$$\hat{\mathbf{S}}^{\text{cw}} = \Phi_{\text{S6}}^{(L)}\left(\mathbf{S}^{\text{cw}}\right), \qquad \hat{\mathbf{S}}^{\text{ccw}} = \Phi_{\text{S6}}^{(L)}\left(\mathbf{S}^{\text{ccw}}\right). \tag{3}$$

The two outputs are then permuted back to the original raster order and reshaped into feature maps

$$\hat{\mathbf{X}}^{\text{cw}} = \text{Permute}_{\pi_{\text{sp}}^{\text{cw}}}^{-1}\left(\hat{\mathbf{S}}^{\text{cw}}\right), \quad \hat{\mathbf{X}}^{\text{ccw}} = \text{Permute}_{\pi_{\text{sp}}^{\text{ccw}}}^{-1}\left(\hat{\mathbf{S}}^{\text{ccw}}\right)$$
$$\mathbf{F}^{\text{cw}} = \text{Unflatten}\left(\hat{\mathbf{X}}^{\text{cw}}\right), \quad \mathbf{F}^{\text{ccw}} = \text{Unflatten}\left(\hat{\mathbf{X}}^{\text{ccw}}\right). \tag{4}$$

Finally, a lightweight learnable direction-balancing gate is used to fuse the two directional features

$$\boldsymbol{\alpha} = \text{Softmax}(\mathbf{w}), \qquad \mathbf{F}_{\text{inter}} = \alpha_1\mathbf{F}^{\text{cw}} + \alpha_2\mathbf{F}^{\text{ccw}} \tag{5}$$

where $\mathbf{w} \in \mathbb{R}^2$ are learnable parameters and $\boldsymbol{\alpha} = [\alpha_1, \alpha_2]$. Although the clockwise and counterclockwise scans share the same periphery-to-center principle, their opposite within-ring visiting orders provide complementary sequential views of the patch. The learnable softmax gate adaptively balances these two directional features during training, rather than imposing a fixed directional preference. This fusion preserves the center-aware information flow while mitigating the direction-dependent bias caused by a single spiral traversal, producing globally regulated features $\mathbf{F}_{\text{inter}}$ for the subsequent local refinement stage.

2) S6 Block (Selective State Space With Discretized SSM): Given a spiral-ordered sequence $\mathbf{S} \in \mathbb{R}^{N\times d}$, each S6 block first generates a content stream and a gating stream

$$\mathbf{S}_{\text{in}}, \mathbf{G}_0 = \text{split}(\mathbf{S}\mathbf{W}_{\text{in}}, 2d), \quad \mathbf{S}_{\text{in}}, \mathbf{G}_0 \in \mathbb{R}^{N\times d}. \tag{6}$$

The content stream is refined by a depthwise 1-D convolution along the sequence dimension

$$\mathbf{X} = \text{SiLU}(\text{DWConv}(\mathbf{S}_{\text{in}})) \in \mathbb{R}^{N\times d}. \tag{7}$$

A linear projection then generates input-dependent selectors

$$\mathbf{T}, \mathbf{B}, \mathbf{C} = \text{split}(\mathbf{X}\mathbf{W}_{\text{sel}}, r, n, n) \tag{8}$$

where $\mathbf{T} \in \mathbb{R}^{N\times r}$ is mapped to per-channel step sizes, $\boldsymbol{\Delta} = \mathbf{T}\mathbf{W}_{\Delta} \in \mathbb{R}^{N\times d}$, and $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{N\times n}$ parameterize the selective SSM update.

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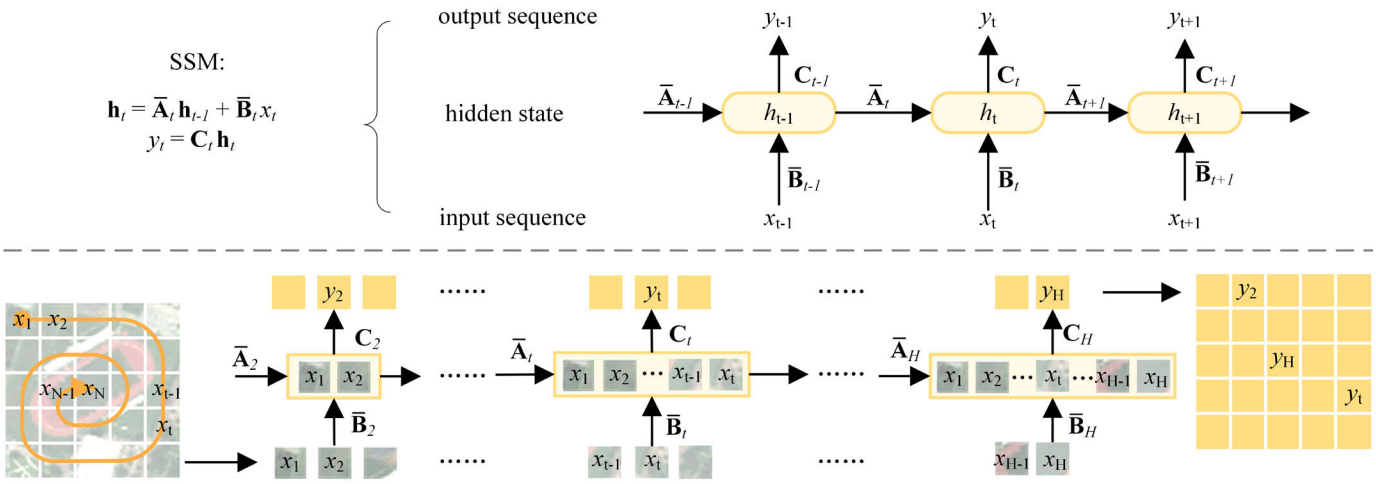


Fig. 5. Context regulation with Spiral-Scan Mamba. (Top) Order-sensitive sequence modeling in mamba via selective state-space dynamics. (Bottom) A periphery-to-center spiral scan converts a 2-D patch into a sequence, so that contextual information is progressively accumulated toward the center token, producing a stabilized global representation for subsequent local refinement.

For each channel $c \in \{1, \ldots, d\}$, the block maintains an SSM state $\mathbf{h}_t^{(c)} \in \mathbb{R}^n$ initialized as $\mathbf{h}_0^{(c)} = \mathbf{0}$. With zero-order-hold discretization, along the scan index $t = 1, \ldots, N$

$$\begin{aligned} \bar{\mathbf{A}}_t^{(c)} &= \exp\left(\Delta_{t,c}\, \mathbf{A}^{(c)}\right) \\ \bar{\mathbf{B}}_t^{(c)} &= \Delta_{t,c}\, \mathbf{B}_t \\ \mathbf{h}_t^{(c)} &= \bar{\mathbf{A}}_t^{(c)} \mathbf{h}_{t-1}^{(c)} + \bar{\mathbf{B}}_t^{(c)}\, X_t^{(c)} \\ y_t^{(c)} &= \mathbf{C}_t^{\top} \mathbf{h}_t^{(c)}. \end{aligned} \tag{9}$$

Stacking all channels yields $\mathbf{Y} \in \mathbb{R}^{N \times d}$. Finally, the gating stream modulates the content output via element-wise multiplication

$$\mathbf{G} = \mathrm{SiLU}\left(\mathbf{G}_0\right), \quad \mathbf{S}' = \left(\mathbf{G} \odot \mathbf{Y}\right) \mathbf{W}_{\mathrm{out}} \in \mathbb{R}^{N \times d}. \tag{10}$$

Stacking L such blocks yields $\Phi_{\mathrm{S6}}^{(L)}(\cdot)$ in (3).

C. Spatial–Spectral Refinement Module and Center-Supervised Patch-Level Prediction

The proposed SSRM refines the globally regulated feature map $\mathbf{F}_{\mathrm{inter}} \in \mathbb{R}^{h \times w \times d}$ using a lightweight two-branch design.

A *spatial* branch emphasizes local structures (e.g., boundaries) via a 3×3 convolution, batch normalization (BN), and nonlinearity

$$\mathbf{F}_{\mathrm{spa}} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times 3}\left(\mathbf{F}_{\mathrm{inter}}\right)\right)\right) \in \mathbb{R}^{h \times w \times d}. \tag{11}$$

In parallel, a *spectral* branch recalibrates channel responses via a 1×1 convolution, layer normalization (LN), and nonlinearity

$$\mathbf{F}_{\mathrm{spe}} = \mathrm{ReLU}\left(\mathrm{LN}\left(\mathrm{Conv}_{1\times 1}\left(\mathbf{F}_{\mathrm{inter}}\right)\right)\right) \in \mathbb{R}^{h \times w \times d}. \tag{12}$$

The refined output is obtained by element-wise fusion

$$\mathbf{F}_{\mathrm{out}} = \mathbf{F}_{\mathrm{spa}} + \mathbf{F}_{\mathrm{spe}} \in \mathbb{R}^{h \times w \times d}. \tag{13}$$

It is worth noting that, although SSRM itself relies on lightweight local operators, it operates on the globally regulated feature map $\mathbf{F}_{\mathrm{inter}}$. As a result, each local refinement implicitly aggregates information that has already incorporated long-range contextual dependencies, thereby effectively enlarging the receptive field of the subsequent CNN-based refinement. This enables the refinement stage to better preserve boundary-sensitive structures and center-consistent semantics, while mitigating the interference from center-irrelevant context under heterogeneous patches.

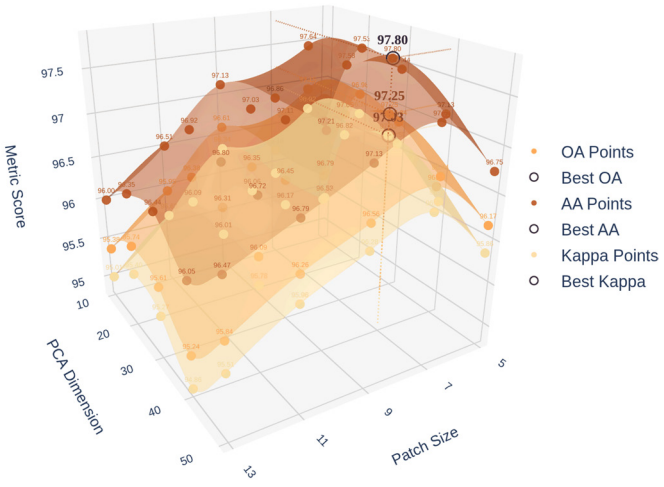


Fig. 6. Hyperparameter analysis of PCA dimensionality and patch size on Houston2013.

Center-Supervised Patch-Level Prediction: Although the supervision is defined on the center pixel, we predict the label from a compact patch descriptor aggregated from $\mathbf{F}_{\mathrm{out}}$. Specifically, we apply global average pooling (GAP) over spatial locations to obtain

$$\mathbf{z} = \mathrm{GAP}\left(\mathbf{F}_{\mathrm{out}}\right), \qquad \hat{\mathbf{y}} = \mathrm{MLP}\left(\mathbf{z}\right) \tag{14}$$

where MLP denotes a multilayer perceptron. In this way, the neighborhood is exploited as auxiliary context, while the training target corresponds to the center pixel.

IV. Experiment

This section validates the proposed center-supervised patch-level HSI–LiDAR classifier and its global-to-local design for mitigating heterogeneous-patch ambiguity. Section IV-A

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TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

NO.	Class(Train/Test)	local neighborhood modeling		within-patch global interaction			hybrid designs(local→global)			global→local
		ENDNet*	S2ENet*	MFT*	HLMamba*	MSFMamba*	S2EFT*	HCTNet*	MHST*	The proposed
1	Healthy grass (20/1231)	95.29	94.72	94.64	97.16	93.1	83.19	**98.05**	97.24	**98.05**
2	Stressed grass (20/1234)	92.06	**99.76**	95.14	94.73	94.89	95.02	98.95	98.87	98.95
3	Synthetic grass (20/677)	98.38	99.85	**100**	97.93	98.38	91.49	99.85	**100**	**100**
4	Trees (20/1224)	94.2	93.3	86.19	96.98	96	98.96	98.77	**99.67**	99.18
5	Soil (20/1222)	99.75	99.92	93.54	84.37	99.18	99.34	99.1	99.84	**100**
6	Water (20/305)	96.39	98.03	97.7	95.41	90.82	93.71	**100**	98.36	**100**
7	Residential (20/1248)	80.13	96.07	92.87	91.67	89.34	92.82	94.87	96.15	**97.92**
8	Commercial (20/1224)	94.28	93.95	**95.83**	86.93	85.46	89.93	88.81	74.67	91.67
9	Road (20/1232)	72.65	**97.08**	83.85	87.99	78.08	82.06	85.39	87.99	91.31
10	Highway (20/1207)	86.5	92.05	95.44	98.26	87.16	66.02	90.47	94.61	**99.75**
11	Railway (20/1215)	85.68	**92.76**	87.41	78.44	91.6	90.04	86.09	90.53	91.77
12	Parking lot 1 (20/1213)	78.24	76.75	85.99	85	86.31	76.08	96.78	92.33	**98.35**
13	Parking lot 2 (20/449)	45.21	**100**	93.32	93.99	90.65	58.95	98	98.44	**100**
14	Tennis court (20/408)	99.75	96.81	**100**	**100**	**100**	94.74	**100**	99.75	**100**
15	Running track (20/640)	99.38	**100**	99.38	**100**	**100**	82.24	**100**	99.22	**100**
	OA(%)	88.05	94.59	92.31	91.45	91.23	86.94	94.72	94.22	**97.25**
	AA(%)	87.86	95.4	93.42	92.59	92.06	86.3	95.68	95.18	**97.8**
	Kappa(%)	87.07	94.16	91.7	90.78	90.52	85.82	94.3	93.75	**97.23**
	Params (M)	0.0883	0.2741	0.2308	0.2745	1.0973	0.1080	0.4529	1.1646	0.3857
	FLOPs (G)	0.0056	0.8522	0.3774	5.7924	3.1202	0.2707	0.5689	7.1836	1.2077

* denotes it is re-implemented by us using the experiment setting provided in its paper. Best results are marked in bold.

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE MUUFL DATASET

NO.	Class(Train/Test)	local neighborhood modeling		within-patch global interaction			hybrid designs(local→global)			global→local
		ENDNet*	S2ENet*	MFT*	HLMamba*	MSFMamba*	S2EFT*	HCTNet*	MHST*	The proposed
1	Trees (20/23226)	87.25	82.76	77.04	74.52	**41.43**	**88.73**	81.96	81.18	85.9
2	Mostly grass (20/4250)	79.53	57.32	**74.96**	52.4	57.55	67.41	70.52	81.41	**83.65**
3	Mixed ground surface (20/6862)	47.64	**74.05**	53.93	40.19	20.74	58.35	58.12	54.01	68.67
4	Dirt and sand (20/1806)	70.54	72.31	68	40.97	40.81	63.46	**59.97**	67.28	**77.96**
5	Road (20/6667)	**87.57**	79.18	73.56	62.46	46.14	74.64	**62.67**	82.96	79.18
6	Water (20/446)	**99.55**	88.79	82.29	65.7	82.06	97.98	82.51	85.87	93.27
7	Building shadow (20/2213)	**87.98**	84.55	65.02	87.35	44.24	76.95	87.62	70.31	**94.71**
8	Building (20/6220)	**93.87**	87.67	82.85	79.49	79.97	92.86	83.73	81.06	89.24
9	Sidewalk (20/1365)	38.32	68.72	60.22	51.5	33.19	**33.85**	50.7	**72.16**	70.48
10	Yellow curb (20/163)	**92.64**	81.6	74.23	61.96	55.21	75.46	74.23	81.6	91.41
11	Cloth panels (20/249)	98.8	**100**	95.98	80.72	93.98	96.79	95.58	91.97	98.8
	OA(%)	80.75	79.23	73.04	66.16	45.64	79.19	74.52	76.85	**82.9**
	AA(%)	80.33	79.72	73.46	63.39	54.12	75.13	73.42	77.25	**84.84**
	Kappa(%)	75.24	73.57	66.53	58.31	36.11	73.07	67.93	70.71	**78.2**
	Params (M)	0.0874	0.1196	0.2145	0.2262	2.8185	0.0965	0.3602	0.8721	0.6317
	FLOPs (G)	0.0056	0.1872	0.1184	3.1546	7.8262	0.0822	0.1522	4.9360	1.1243

* denotes it is re-implemented by us using the experiment setting provided in its paper. Best results are marked in bold.

describes the datasets. Section IV-B introduces the experimental settings, evaluation metrics, and compared methods. Section IV-C reports the quantitative and qualitative comparisons, along with the efficiency analysis. Section IV-D presents ablation studies. Section IV-E further discusses the superiority of the global-to-local refinement and the effectiveness of the proposed Spiral Scan.

A. Datasets

We evaluate on three widely-used HSI–LiDAR benchmarks: Houston2013, MUUFL, and Augsburg. All datasets provide co-registered HSI and LiDAR measurements with pixel-wise land-cover labels, enabling patch-based sampling where each neighborhood is supervised by its center pixel. The class-wise sample distributions for training and testing are summarized in Tables I–III.

1) Houston2013: Houston2013 was released for the IEEE GRSS Data Fusion Contest and covers the University of Houston campus and surrounding urban areas. The HSI image contains 144 bands (364–1046 nm), and the LiDAR modality provides a single-band elevation map. This dataset includes 15 urban land-cover and land-use classes, featuring strong class mixing in local neighborhoods due to complex man-made structures and vegetation interleaving.

2) MUUFL: The MUUFL Gulfport dataset covers the University of Southern Mississippi Gulf Park campus [46]. The HSI modality contains 64 denoised bands spanning 375–1050 nm, and the LiDAR modality provides elevation information. Compared with the other datasets, MUUFL is

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TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE AUGSBURG DATASET

NO.	Class(Train/Test)	local neighborhood modeling		within-patch global interaction			hybrid designs(local→global)			global→local
		ENDNet*	S2ENet*	MFT*	HLMamba*	MSFMamba*	S2EFT*	HCTNet*	MHST*	The proposed
1	Forest (20/13487)	92.69	78.56	82.16	**80.44**	87.88	93.68	85.1	89.89	**93.82**
2	Residential Area (20/30309)	62.12	74.79	71.19	77.83	82.48	69.12	**83.63**	69.14	**83.92**
3	Industrial Area(20/3831)	46.38	7.7	50.87	59.7	29.39	24.54	**60.85**	45.5	**67.08**
4	Low Plants(20/26837)	62.46	82.58	**71.05**	69.17	75.09	46.12	61.87	55.46	**85.72**
5	Allotment (20/555)	20	76.04	64.86	65.77	73.51	61.08	69.19	**90.45**	**96.58**
6	Commercial Area(20/1625)	23.14	**92.18**	53.23	52.37	60.92	**54.58**	37.72	46.34	59.26
7	Water(20/1510)	72.19	51.52	67.28	58.94	55.63	51.19	68.15	**69.07**	**74.3**
	OA(%)	65.83	74.75	71.54	73.44	77.24	62.57	73.94	66.54	**84.81**
	AA(%)	54.14	66.2	65.81	66.32	66.42	57.19	66.64	66.55	**80.1**
	Kappa(%)	55.14	66.14	62.61	64.93	69.33	49.63	65.29	56.95	**79.29**
	Params (M)	0.0878	0.1540	0.2302	0.2314	0.9043	0.1029	0.4521	1.2986	0.6317
	FLOPs (G)	0.0056	0.2425	0.2702	3.4849	2.9899	0.2615	0.3141	8.2208	1.1243

* denotes it is re-implemented by us using the experiment setting provided in its paper. Best results are marked in bold.

relatively high-resolution, where fine-grained local structures (e.g., narrow roads and shadows) are prominent and easily induce within-patch heterogeneity.

3) Augsburg: The Augsburg dataset captures the Augsburg area in Germany [47]. It provides an HSI image (180 bands, 400–2500 nm) and a single-band LiDAR modality, with seven urban land-cover classes. Due to its coarse spatial resolution, local neighborhoods often contain mixed materials and blurred boundaries, making center-pixel discrimination under heterogeneous patches particularly challenging.

B. Experimental Setup

1) Implementation Details: All experiments are implemented in PyTorch and trained on a single NVIDIA RTX A6000 GPU. We optimize the network using AdamW with cross-entropy loss. The batch size is 64, and the model is trained for 200 epochs. Unless otherwise stated, all reported results are obtained under the same training and testing splits for fair comparison.

a) Preprocessing and hyperparameter settings: For each dataset, the HSI cube is first reduced by PCA to a compact spectral subspace, which alleviates spectral redundancy and improves training stability. Both modalities are normalized independently. We then extract an HSI–LiDAR patch centered at each labeled pixel to form a training or testing sample. We tune two key hyperparameters: 1) PCA dimensionality for HSI and 2) patch size. These two factors jointly control the tradeoff between spectral compactness and contextual coverage.

Following the analysis in Section IV-B4, we use (PCA, patch) = (40, 7) for Houston2013, (10, 5) for MUUFL, and (40, 5) for Augsburg (unless explicitly varied).

2) Evaluation Metrics: We report per-class accuracy as well as overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. For qualitative evaluation, we visualize classification maps to assess spatial coherence and boundary quality, which are closely related to heterogeneous-patch ambiguity. We also report computational complexity in terms of parameters and FLOPs to assess the accuracy–efficiency tradeoff (Tables I–III).

In addition, to quantify heterogeneous-patch ambiguity, we define a center-label inconsistency ratio for each test patch. Let P_c denote an $h \times w$ patch centered at pixel c, and let

$$\Omega_c = \left\{(i, j) \in P_c \,\middle|\, y_{i,j} \neq 0\right\} \tag{15}$$

be the set of labeled pixels within the patch. Since the adopted HSI–LiDAR benchmarks are sparsely annotated, unlabeled pixels are excluded from the ambiguity calculation. The inconsistency ratio is defined as

$$R_{\text{inc}}(P_c) = \frac{1}{|\Omega_c|} \sum_{(i,j)\in\Omega_c} \mathbf{1}\left(y_{i,j} \neq y_c\right) \tag{16}$$

where y_c is the center-pixel label. A larger $R_{\text{inc}}(P_c)$ indicates that more labeled contextual pixels are inconsistent with the center target, corresponding to stronger heterogeneous-patch ambiguity.

Based on $R_{\text{inc}}(P_c)$, test samples are divided into three ambiguity levels. Samples with $R_{\text{inc}}(P_c) = 0$ are assigned to the low-ambiguity group, while samples with $R_{\text{inc}}(P_c) > 0$ are further split into medium- and high-ambiguity groups according to the median of the nonzero inconsistency ratios. For each ambiguity group, AA is computed in the same way as in the standard evaluation, i.e., by averaging per-class accuracies.

3) Compared Methods: To reflect the research landscape summarized in Section II, we compare with representative multimodal methods from three categories: 1) *local neighborhood modeling* (CNN-based modeling), e.g., ENDNet [48] and S2ENet [11]; 2) *within-patch global interaction* (ViT/Mamba-style modeling), including a ViT-based baseline (MFT [12]) and Mamba-style baselines (e.g., HLMamba [49], MSFMamba [50]); and 3) *hybrid designs* that combine global and local modeling, e.g., S2EFT [51], HCTNet [52], and MHST [41]. We follow official implementations whenever available; otherwise, we re-implement methods under the reported settings and adapt them to our experimental setup.

4) Hyperparameter Analysis: We conduct hyperparameter analysis from two aspects: input-related settings, including PCA dimensionality and patch size, and architecture-related settings, including the number of Mamba layers L and the

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TABLE IV
SENSITIVITY ANALYSIS OF MAMBA LAYERS L AND EMBEDDING DIMENSION d ON THE HOUSTON2013 DATASET. THE DEFAULT SETTING IS MARKED BY †

Setting	Value	OA(%)	AA(%)	Kappa(%)
Effect of the number of Mamba layers L				
L	1	96.81	97.43	96.55
L	3	95.64	96.50	95.29
L	**5**†	**97.25**	**97.80**	**97.23**
L	7	96.94	97.49	96.69
L	9	96.38	97.07	96.09
Effect of the embedding dimension d				
d	32	96.80	97.41	96.54
d	**64**†	**97.25**	**97.80**	**97.23**
d	128	96.65	97.28	96.38
d	256	95.19	96.01	94.80

embedding dimension d. Houston2013 is used as the representative benchmark for detailed analysis because it contains diverse land-cover categories and complex spatial structures. Comparable tuning was also conducted on MUUFL and Augsburg following the same protocol, and consistent trends are observed. The final selected settings for all datasets are reported in Section IV-B.

For input-related hyperparameters, Fig. 6 shows the influence of PCA dimensionality and patch size. Overall, patch size has a more pronounced influence than PCA dimensionality. This is consistent with the center-supervised patch-based setting, where the neighborhood size directly determines the amount of contextual evidence available for the center-pixel prediction. Moderate patch sizes, such as (7×7), provide a better balance between contextual coverage and heterogeneous clutter, while PCA mainly serves to reduce spectral redundancy and improve feature compactness.

We further analyze the architecture-related hyperparameters of the Mamba-based global regulation module. Specifically, we vary the number of Mamba layers L and the embedding dimension d while keeping other settings unchanged. The results are reported in Table IV.

As shown in Table IV, the proposed method obtains the best performance when $L = 5$ and $d = 64$. The results under nearby settings, such as $L = 7$ and $d = 32$, remain competitive, indicating that the model is not overly sensitive to moderate changes in these architectural hyperparameters. However, excessively increasing the depth or width does not lead to further improvement and may even degrade performance. This suggests that simply enlarging the Mamba module is unnecessary for the center-supervised small-sample setting and may introduce redundant representations. Therefore, we adopt $L = 5$ and $d = 64$ as the default configuration, which provides strong performance while keeping the global regulation module compact.

C. Comparison With SOTA Methods

1) Quantitative Results: Across all three benchmarks, our method achieves the best overall performance, demonstrating robust generalization under varying spatial resolutions and heterogeneous neighborhoods (Tables I–III). More importantly, the gains are consistent across the three representative baseline groups discussed in Section II: 1) *local neighborhood modeling* (CNN-based fusion); 2) *within-patch global interaction* (ViT/Mamba-style modeling); and 3) *hybrid designs* that combine global and local modeling.

a) Houston2013: Our method achieves OA/AA/Kappa of 97.25%/97.80%/97.23% (Table I). Compared with the strongest *hybrid* baseline (HCTNet), it improves OA, AA, and Kappa by 2.53%, 2.12%, and 2.73%, respectively, and yields the best accuracy in 10 out of 15 classes. In addition, it consistently surpasses *local* CNN-based fusion methods (ENDNet, S2ENet) and *global* within-patch interaction models (MFT and Mamba-style baselines), indicating that explicitly organizing context in a global-to-local manner is effective for suppressing semantically irrelevant interference under heterogeneous neighborhoods.

b) MUUFL: Our method reaches OA/AA/Kappa of 82.90%/84.84%/78.20% (Table II). MUUFL has high spatial resolution where fine-grained local cues dominate, making *local* CNN-based fusion and some *hybrid* designs generally competitive. In contrast, purely *global* interaction models (ViT/Mamba-style) may suffer from weakened locality. Still, our method remains the best, suggesting that the proposed global regulation is not a generic long-range modeling component; rather, it provides center-relevant context organization that complements strong local evidence, which is beneficial even in high-resolution scenarios.

c) Augsburg: Our method significantly outperforms all baselines with OA/AA/Kappa of 84.81%/80.10%/79.29% (Table III). Given the coarser spatial resolution and stronger within-patch mixing, relying solely on *local* neighborhood modeling becomes less reliable, while robust *global* context organization becomes more critical. The strong gains over both *global* baselines (MFT and Mamba-style models) and *hybrid* baselines verify that our center-aware global regulation (Spiral Scan) plus local refinement (SSRM) better matches the center-supervised patch setting, alleviating heterogeneous-patch ambiguity.

Overall, the consistent improvements over the three baseline families validate our core design: under center-supervised patch-level prediction, it is crucial to: 1) regulate within-patch context in a center-aware manner (Spiral Scan) and 2) refine locally to recover boundary-sensitive and class-consistent cues (SSRM), rather than relying on only local fusion, only global interaction, or loosely coupled hybrids.

d) Remaining challenging cases: Several categories with narrow spatial support or strong contextual mixing remain challenging, such as roads/sidewalks on MUUFL and built-up areas on Augsburg. For these categories, the center pixel is often surrounded by visually or semantically competing pixels, so the center-consistent evidence within a patch can be insufficient or spatially fragmented. Similar class-wise fluctuations are also observed for representative local, global, and hybrid baselines, suggesting that these categories are generally difficult because their local patches often contain mixed or competing contextual pixels.

2) Qualitative Results: The qualitative results on the three benchmark datasets are shown in Figs. 7–9, providing visual

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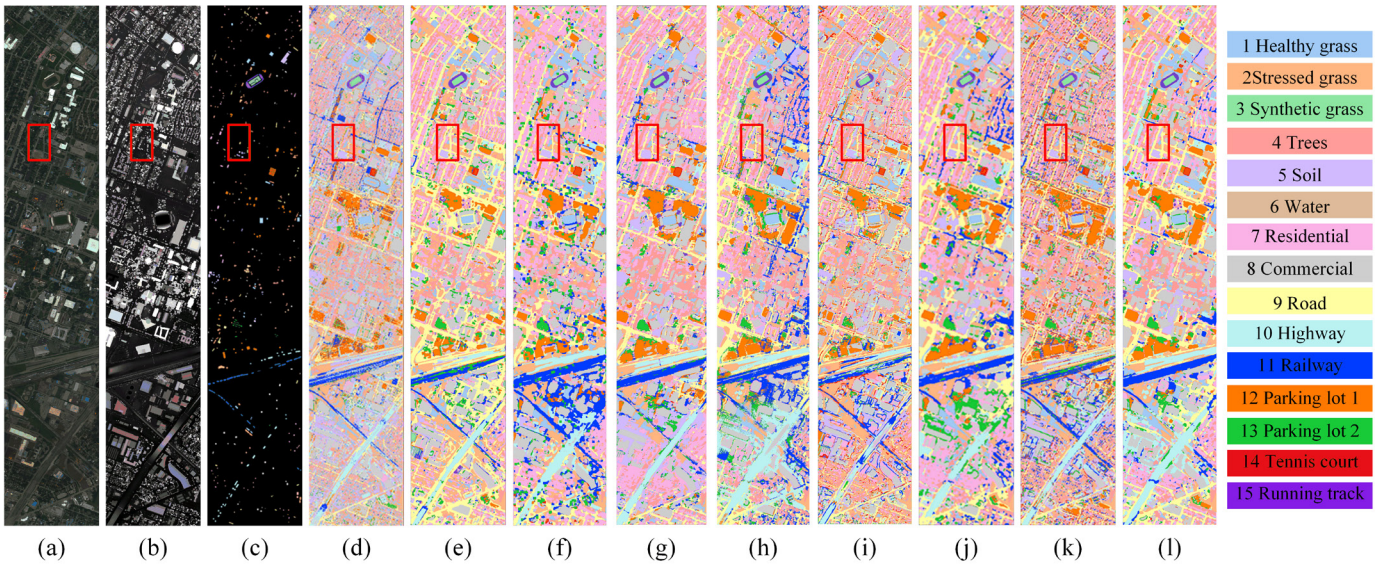


Fig. 7. Classification maps by different methods on the Houston2013 dataset. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground-truth map. (d) EndNet. (e) S2ENet. (f) MFT. (g) HLMamba. (h) MSFMamba. (i) S2EFT. (j) HCTNet. (k) MHST. (l) Proposed.

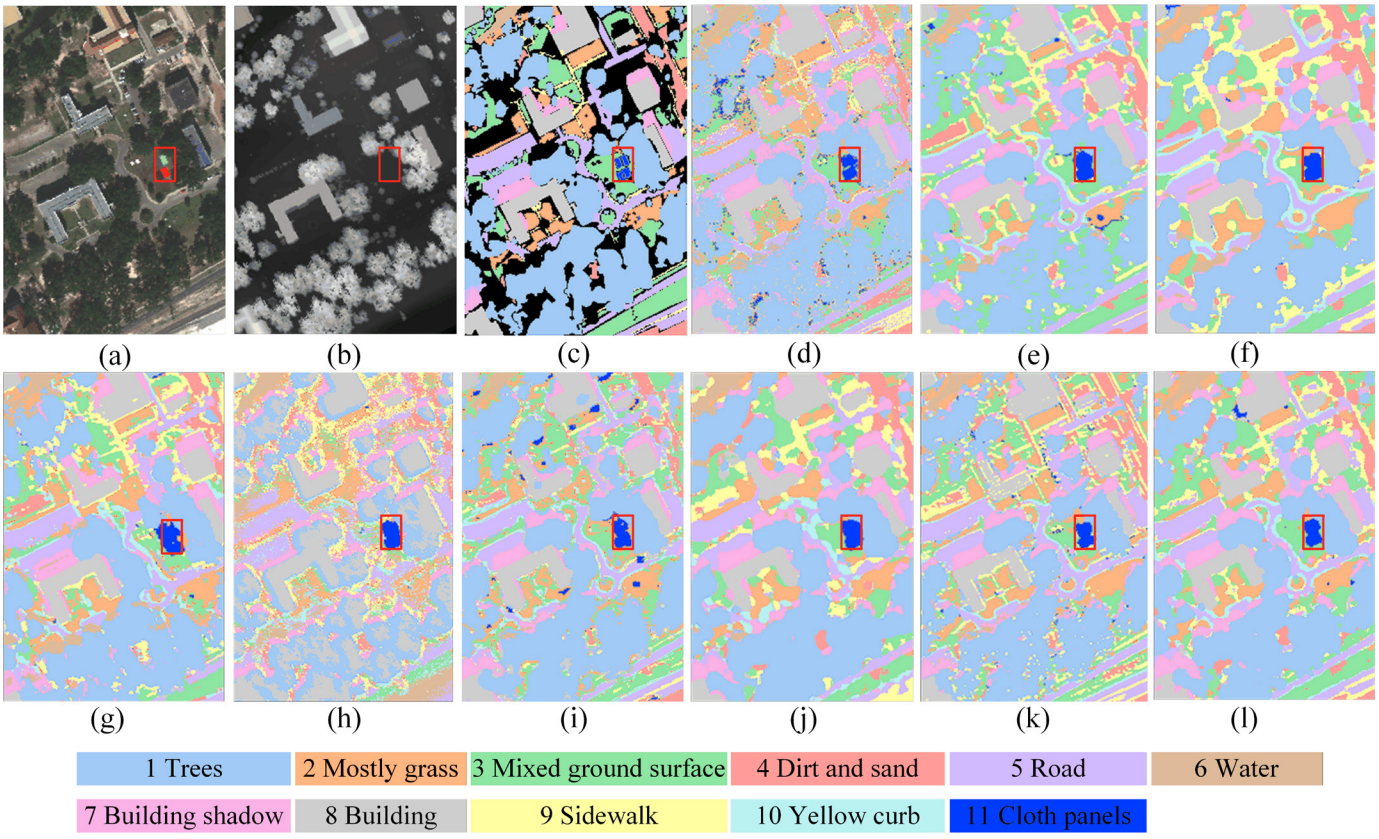


Fig. 8. Classification maps by different methods on the MUUFL dataset. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground-truth map. (d) EndNet. (e) S2ENet. (f) MFT. (g) HLMamba. (h) MSFMamba. (i) S2EFT. (j) HCTNet. (k) MHST. (l) Proposed.

evidence that complements the quantitative evaluation. Across all datasets, existing methods exhibit common limitations when handling heterogeneous neighborhoods, including fragmented predictions and blurred class boundaries, particularly in areas where multiple land-cover types coexist.

As illustrated in Fig. 7, such issues are prominent in densely built urban scenes, where local neighborhood modeling methods produce noticeable salt-and-pepper noise and discontinuous linear structures, such as roads and railways. In Fig. 8, scenes with frequent class transitions further expose the limitations of generic global interaction, where indiscriminate contextual mixing leads to boundary blurring or over-smoothed regions. Similarly, Fig. 9 shows that conventional hybrid designs may still suffer from frag-

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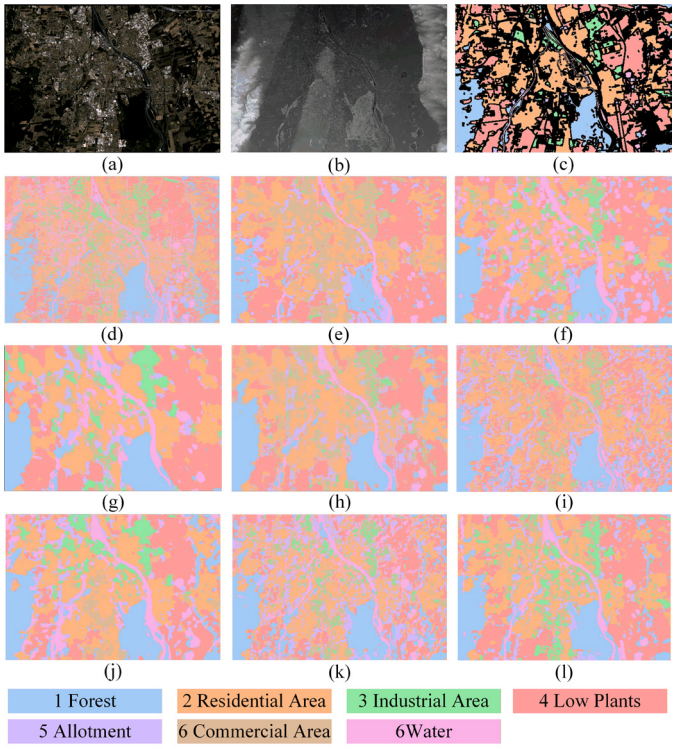


Fig. 9. Classification maps by different methods on the Augsburg dataset. (a) Pseudo-color image for HSI. (b) LiDAR-based DSM. (c) Ground-truth map. (d) EndNet. (e) S2ENet. (f) MFT. (g) HLMamba. (h) MSFMamba. (i) S2EFT. (j) HCTNet. (k) MHST. (l) Proposed.

mented predictions near land–water boundaries or within large homogeneous regions due to insufficient regulation of heterogeneous context.

By contrast, the proposed method consistently yields more spatially coherent classification maps across all three datasets. The center-aware global regulation stage stabilizes patch-level representations by suppressing interference from semantically irrelevant neighboring regions, while the subsequent spatial–spectral refinement restores boundary-sensitive details under the globally regulated context. As a result, the proposed framework produces smoother interiors, clearer boundaries, and more continuous structures, as exemplified by the highlighted regions, demonstrating improved robustness to heterogeneous neighborhoods.

3) Computational Efficiency: We assess model complexity using parameter count and FLOPs on all three benchmarks (Tables I–III). The reported complexity values are obtained under the corresponding experimental setting of each dataset, consistent with the accuracy comparisons in the same table.

Across the three datasets, the proposed method achieves the best OA while maintaining a moderate computational cost. Compared with Mamba-style baselines such as HLMamba and MSFMamba, the proposed method obtains higher accuracy with lower or comparable FLOPs on the corresponding benchmarks. Compared with hybrid designs, our method is not always the lightest model, but it provides a stronger accuracy–complexity balance. These results support the claimed favorable accuracy–efficiency tradeoff under the adopted experimental settings.

TABLE V
AA (%) ON THE MIXED TEST SET AND DIFFERENT HETEROGENEOUS-PATCH AMBIGUITY LEVELS

Method	Mix	Low	Medium	High
Houston2013				
S2ENet	95.40	99.21	96.15	**95.83**
MFT	93.42	98.72	**98.08**	**95.83**
HCTNet	95.68	97.77	94.76	94.36
The proposed	**97.80**	**99.27**	**98.08**	**95.83**
MUUFL				
S2ENet	79.72	93.45	89.28	70.17
MFT	73.46	92.82	86.87	69.62
HCTNet	73.42	85.56	80.40	63.40
The proposed	**84.84**	**96.71**	**91.60**	**71.74**
Augsburg				
S2ENet	66.20	80.84	70.03	62.03
MFT	65.81	78.65	**73.58**	59.62
HCTNet	66.64	70.72	62.01	52.54
The proposed	**80.10**	**83.64**	70.34	**62.54**

Notes. Mix denotes the standard test set used in Tables II–IV. Low, Medium, and High are divided according to $R_{\text{inc}}(P_c)$.

4) Heterogeneous-Patch Ambiguity-Level Analysis: The quantitative comparisons in Tables I–III report the standard results on the mixed test set, where samples with different degrees of heterogeneous-patch ambiguity are evaluated together. To further analyze this factor, we divide the same test samples into low-, medium-, and high-ambiguity subsets according to the center-label inconsistency ratio $R_{\text{inc}}(P_c)$, and report AA on both the mixed test set and each subset in Table V. For representative comparison, we select one strong baseline from each method family: S2ENet, MFT, and HCTNet.

Several observations can be made. First, AA generally decreases from low to medium and high ambiguity across the three datasets, indicating that stronger heterogeneous-patch ambiguity increases classification difficulty. Second, this trend appears for both the proposed method and representative baselines, suggesting that $R_{\text{inc}}(P_c)$ reflects a common difficulty factor in center-supervised patch-based classification. Third, the proposed method achieves the best or tied-best results in most columns, especially on MUUFL and Augsburg with more pronounced within-patch mixing.

These results quantitatively support our problem formulation and further validate the robustness of the proposed center-aware global-to-local refinement strategy under heterogeneous patches.

D. Ablation Study

We ablate the key components of the proposed framework on the Houston2013 dataset under a unified training setting. As shown in Table VI, the complete configuration, i.e., CCA + M + Spiral + SSRM, achieves the best performance.

1) Effect of Cross-Modal Channel Alignment: We first evaluate the proposed CCA from two aspects: fusion placement and early fusion strategy. For fusion placement, late fusion decreases OA, AA, and Kappa by 2.21%, 1.79%, and 2.59%, respectively. This indicates that spectral–elevation complementarity should be available during Mamba-based

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TABLE VI

ABLATION STUDY OF THE PROPOSED MODULES ON THE HOUSTON2013 DATASET. THE REFERENCE IS **CCA + M + SPIRAL + SSRM**. ΔOA DENOTES THE OA GAP TO THE REFERENCE

Fusion	Global	Local	OA(%)	ΔOA	AA(%)	Kappa(%)
Effect of fusion placement						
Late CCA	Sep. M+Spiral	SSRM	95.04	-2.21	96.01	94.64
Effect of early fusion strategy						
DConcat	M+Spiral	SSRM	94.83	-2.42	95.76	94.41
ConvF	M+Spiral	SSRM	96.67	-0.58	97.28	96.40
AttF	M+Spiral	SSRM	95.46	-1.79	96.30	95.10
Effect of global module						
CCA	*T*	SSRM	94.99	-2.26	95.94	94.59
Effect of local module branches						
CCA	M+Spiral	Spatial-only	96.52	-0.73	97.18	96.24
CCA	M+Spiral	Spectral-only	96.88	-0.37	97.40	96.62
Reference						
CCA	**M+Spiral**	**SSRM**	**97.25**	**0.00**	**97.80**	**97.23**

Notes. CCA: proposed cross-modal channel alignment; Late CCA: CCA applied after separate global modeling of HSI and LiDAR; Sep. M+Spiral: separate Mamba with Spiral Scan branches for HSI and LiDAR; DConcat: direct concatenation; ConvF: convolution-based fusion; AttF: attention-based fusion implemented following the cross-modal attention idea [21]; *T*: standard ViT; M+Spiral: Mamba with Spiral Scan; Spatial-only/Spectral-only: single-branch variants of SSRM; SSRM: full spatial–spectral refinement module.

TABLE VII

COMPARISON OF LOCAL AND GLOBAL HYBRID STRATEGIES WITH IDENTICAL MODULES ON THE HOUSTON2013 DATASET. THE REFERENCE IS **GLOBAL → LOCAL (M + SPIRAL–SSRM)**. ΔOA DENOTES THE GAP TO THE REFERENCE (HIGHER IS BETTER)

structure		OA(%)	ΔOA	AA(%)	Kappa(%)
design	realization				
Onlyglobal	Only M+Spiral	96.37	-0.88	97.07	96.08
Onlylocal	Only SSRM	96.84	-0.41	97.45	96.58
local&global	SSRM & M+Spiral	96.64	-0.61	97.22	96.37
local→global	SSRM→M+Spiral	96.88	-0.37	97.49	96.63
global→local	**M+Spiral→SSRM**	**97.25**	**0.00**	**97.80**	**97.23**

Notes. All variants share the same modules; only the hybrid order differs. "M+Spiral" denotes Mamba with Spiral Scan; "SSRM" denotes the spectral-spatial refinement module.

global regulation. If HSI and LiDAR are processed separately and fused only after global modeling, the global regulation stage cannot exploit cross-modal interactions during center-aware context aggregation.

For early fusion strategies, direct concatenation, convolution-based fusion, and attention-based fusion are all inferior to the proposed CCA. Direct concatenation suffers from unbalanced HSI–LiDAR channel distributions, while convolution-based fusion mainly performs local mixing without explicit modality-specific channel alignment. Attention-based fusion [21] can adaptively couple modalities, but it may introduce stronger data-dependent modulation and is less stable under heterogeneous patches. In contrast, CCA first aligns HSI and LiDAR with modality-specific MLPs, then performs cross-modal channel mixing and SE recalibration, thereby providing more balanced multimodal tokens for subsequent Mamba-based center-aware regulation.

2) Effect of Center-Aware Global Regulation: We evaluate the global module by replacing M + Spiral with a standard ViT encoder with the same number of layers and embedding dimension. The ViT variant decreases OA by 2.26%, indicating that the proposed Mamba-based global regulation with Spiral Scan is more suitable for center-supervised patch classification. The order-sensitive SSM recurrence enables structured context accumulation, and the periphery-to-center scan further aligns the global regulation process with the center-pixel prediction objective.

3) Effect of Local Refinement: We further ablate SSRM by enabling only the spatial or only the spectral branch. Both single-branch variants are inferior to the full SSRM. Compared with the complete SSRM, the spatial-only and spectral-only variants decrease OA by 0.73% and 0.37%, respectively. These results verify that spatial boundary refinement and spectral channel recalibration are complementary for recovering discriminative local details within the globally regulated feature space.

E. Discussion

1) Why Global-to-Local Works Better: Table VII shows that, under the same core modules, the sequential *global-to-local* organization consistently achieves the best overall performance, outperforming both the reversed order and the parallel hybrid. This result suggests that the observed gain is not simply due to the presence of both global and local components, but is closely related to how contextual information is organized and processed across stages.

To provide additional insight, Fig. 10 presents t-SNE visualizations of the classifier-input features under different global–local organization strategies. Overall, the resulting feature distributions exhibit comparable cluster structures across the hybrid configurations, indicating that all variants are capable of learning meaningful semantic representations. Nevertheless, the global-to-local order [Fig. 10(a)] tends to yield slightly more coherent feature groupings for certain categories, while the reversed local-to-global order [Fig. 10(c)] and the parallel design [Fig. 10(b)] occasionally show increased dispersion or overlap among semantically related classes. When only local or only global modeling is employed [Fig. 10(d) and (e)], the feature distributions become less structured, reflecting insufficient contextual integration.

These observations, together with the quantitative results, indicate that placing the global encoder before local refinement provides a more suitable feature basis for center-pixel classification. By first regulating patch-level context at a coarse semantic level, the model reduces the influence of heterogeneous or competing regions, after which local refinement can focus on enhancing boundary-sensitive structures and fine-grained spatial–spectral cues within a stabilized feature space.

This interpretation is further supported by the feature-activation visualizations in Fig. 11. For the same input patches shown in Fig. 11(a), the global-to-local configuration [Fig. 11(c)] produces more spatially concentrated and class-consistent activation patterns, whereas the local-to-global order [Fig. 11(b)] results in relatively more dispersed responses.

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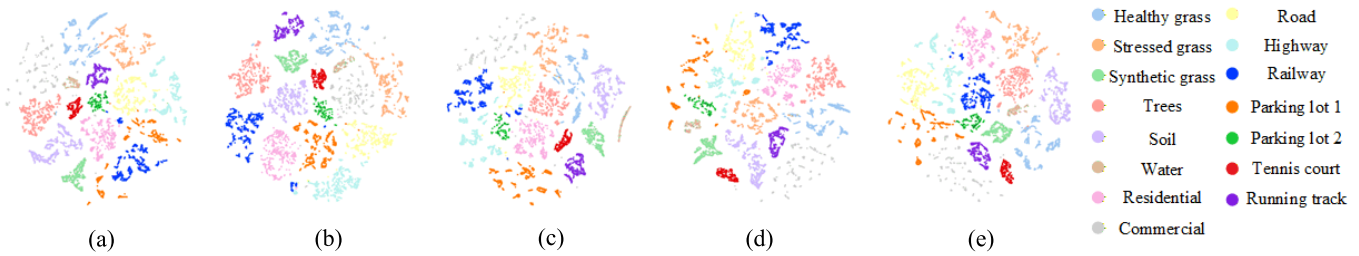


Fig. 10. t-SNE visualization of feature distributions under different global–local organization strategies. The plots show the classifier-input features extracted by different local–global structure variants, where the global component is implemented by Mamba with Spiral Scan and the local component by SSRM, highlighting how a global-to-local ordering improves feature separability. (a) Global → local. (b) Local&global. (c) Local → global. (d) Only local. (e) Only global.

TABLE VIII

COMPARISON OF DIFFERENT SCAN STRATEGIES ON THE HOUSTON2013 DATASET. THE REFERENCE IS SPIRAL SCAN. ΔOA DENOTES THE OA GAP TO THE REFERENCE

Category	Scan Strategy	Scan Paths	OA(%)	ΔOA	AA(%)	Kappa(%)	FLOPs(G)
Generic Vision Mamba scans	Vim Scan	2	96.75	-0.50	97.38	96.48	1.2077
	FRR Scan	3	96.70	-0.55	97.33	96.43	1.2788
	VMamba Scan	4	97.01	-0.24	97.58	96.76	1.3499
	Omnidirectional Scan	8	96.51	-0.74	97.19	96.23	1.6344
Controlled center-biased variants	Distance-Based Ordering	1	96.98	-0.27	97.57	96.73	1.1453
	Radial Scan	1	96.75	-0.50	97.36	96.48	1.1453
Proposed	Spiral Scan	2	**97.25**	0.00	**97.80**	**97.23**	1.2077

Note: All variants use the same network architecture and differ only in the scan order or scan paths.

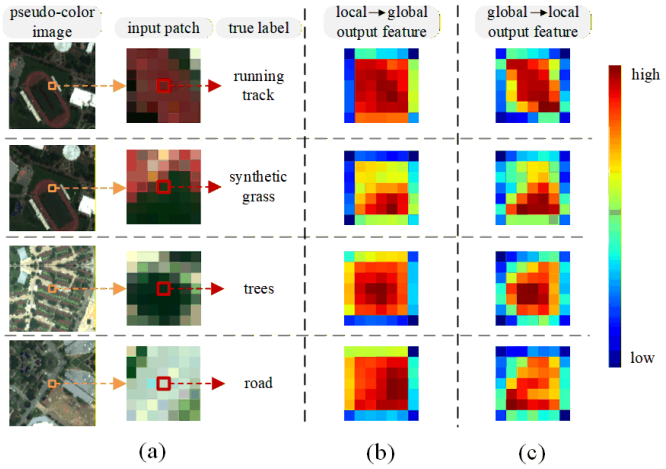


Fig. 11. Feature-activation comparison between local → global and global → local organizations. Global → local produces more concentrated and center-aligned activations than local → global under identical input patches. Heatmaps are obtained via channel-wise averaging and per-sample min–max normalization [53], [54], [55]. (a) input patches, (b) output features of the local-to-global organization, and (c) output features of the global-to-local organization.

Since these heatmaps are derived directly from the final feature tensor via channel-wise averaging and per-sample min–max normalization, the observed differences suggest that the global-to-local ordering facilitates more focused and structured feature responses, which is beneficial for center-supervised patch-based classification.

2) Why Spiral Scan Is Effective for Center-Supervised Classification: Table VIII compares Spiral Scan with representative Vision Mamba scans and controlled center-biased variants on Houston2013. The former includes generic scan patterns such as Vim, FRR, VMamba, and omnidirectional

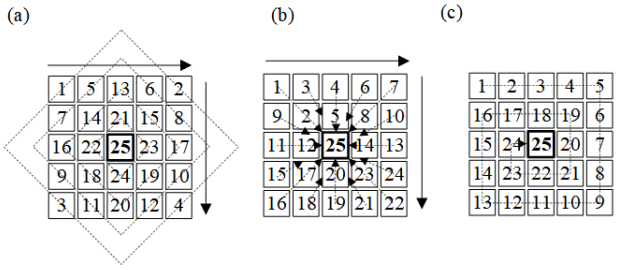


Fig. 12. Illustration of center-biased scan variants. Distance-Based Ordering sorts pixels by Manhattan distance to the center with row-major tie-breaking. Radial Scan traverses boundary-to-center rays. Spiral Scan performs ring-wise periphery-to-center traversal; only the clockwise path is shown, while the counterclockwise path uses the opposite within-ring direction. (a) Distance-Based Ordering. (b) Radial Scan. (c) Spiral Scan (proposed, clockwise shown).

scans, while the latter examines whether the gain of Spiral Scan simply comes from center-biased ordering. Specifically, Distance-Based Ordering and Radial Scan are used as controlled baselines, as illustrated in Fig. 12. They share the periphery-to-center principle with Spiral Scan but differ in serialization patterns.

As shown in Table VIII, Spiral Scan achieves the best overall performance with only two scan paths and moderate computational cost. It outperforms Vim, FRR, VMamba, and omnidirectional scans by 0.50%, 0.55%, 0.24%, and 0.74% in OA, respectively. Although omnidirectional scanning uses more paths, it yields lower accuracy and higher FLOPs than Spiral Scan, indicating that increasing scan directions alone does not necessarily benefit center-supervised patch classification. Instead, the scan order should match the center-pixel prediction objective.

The controlled variants show that Spiral Scan is not better simply because the center pixel is placed near the sequence end. Compared with Distance-Based Ordering, Spi-

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ral Scan improves OA, AA, and Kappa by 0.27%, 0.23%, and 0.30%, respectively; compared with Radial Scan, the gains are 0.50%, 0.44%, and 0.55%. In Mamba, tokens are processed sequentially, so the scan order also defines how contextual information is passed along the hidden state. This is important because abrupt jumps in the sequence may force the hidden state to mix evidence from unrelated regions, whereas spatially neighboring tokens usually provide more compatible contextual cues.

Distance-Based Ordering is center-biased, but its row-major tie-breaking may jump between spatially separated pixels with the same distance to the center. Radial Scan also moves from the boundary to the center, but switching between rays may break the continuity of sequential propagation. By contrast, Spiral Scan moves toward the center while keeping neighboring tokens spatially connected along each ring. Therefore, its advantage over Distance-Based Ordering and Radial Scan comes from combining center-oriented accumulation with spatially coherent state propagation, rather than from periphery-to-center ordering alone.

We further analyze the learned fusion weights of the dual-direction Spiral Scan. Let α_{cw} and α_{ccw} denote the softmax-normalized weights of the clockwise and counterclockwise paths, respectively,

$$\alpha_{\mathrm{cw}}, \alpha_{\mathrm{ccw}} = \mathrm{Softmax}\left(w_{\mathrm{cw}}, w_{\mathrm{ccw}}\right). \tag{17}$$

On Houston2013, the learned weights are $\alpha_{\mathrm{cw}} = 0.5163$ and $\alpha_{\mathrm{ccw}} = 0.4837$, showing a balanced distribution rather than a collapse to a single direction. This suggests that the two scans provide complementary sequential views, while the learnable gate acts as a direction-balancing mechanism.

Overall, Spiral Scan offers a compact and effective context stream for center-supervised HSI–LiDAR classification. By combining periphery-to-center accumulation, spatially coherent state propagation, and dual-direction balanced fusion, it helps Mamba consolidate center-relevant evidence while reducing interference from semantically distant regions under heterogeneous neighborhoods.

V. Conclusion

In this work, we revisited patch-based HSI–LiDAR classification from the perspective of center-supervised learning and identified *heterogeneous-patch ambiguity*, arising from mixed semantics and uneven contextual influence within local neighborhoods, as a fundamental challenge that degrades patch-level representations used for center-pixel prediction. Motivated by this observation, we proposed a center-aware global-to-local refinement framework that first regulates within-patch contextual information at a global stage and then refines discriminative spatial–spectral details through localized enhancement.

Extensive experiments on three benchmark datasets demonstrate that the proposed framework consistently outperforms state-of-the-art methods across different spatial resolutions and heterogeneity levels, while maintaining a favorable accuracy–efficiency trade-off. Further ablation and analysis confirm that: 1) center-aware context regulation is critical for stabilizing patch-level representations under heterogeneous neighborhoods; 2) Spiral Scan provides an effective and structured information accumulation order that facilitates center-relevant semantic consolidation; and 3) localized spatial–spectral refinement complements global regulation by recovering boundary-sensitive details that are decisive for center-pixel classification. Future work will focus on further advancing the global-to-local modeling paradigm, including adaptive designs for context regulation, as well as more flexible global–local interaction schemes within center-supervised patch-based frameworks.

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